FPGA Tick-to-Trade – Implementation Results

June 2026

Vivado 2025.2 | xcu200-fsgd2104-2-e (UltraScale+) | 200 MHz target

Executive Summary

Run	Design	WNS (ns)	Failing eps	Status
B0	`tick_to_trade_top`	−2.373	41,206	**FAIL** route-dominated FF array
B1	`tick_to_trade_top`	−4.563	3,612	**FAIL** combinational chain
M1	`multi_symbol_top`	−5.362	10,899	**FAIL** chain + 4-way sym mux
B2	`tick_to_trade_top`	−2.217	87	**FAIL** I/O artifact; datapath closed
B3	`tick_to_trade_top`	—	—	**PENDING** RTL done; synth not run
B4	`tick_to_trade_top`	−0.578	47	**FAIL** FF→OBUF only; logic **PASS**
B5	`tick_to_trade_top`	**+0.105**	**0**	**PASS 200 MHz CLOSED**
M2	`multi_symbol_top`	*TBD*	*TBD*	**PENDING** XDC updated; **re-run needed**

Key finding: `tick_to_trade_top` closes at 200 MHz (B5, WNS = +0.105 ns, 0 failing endpoints, measured 2026-06-16). The trading datapath (ITCH parse → BRAM book → strategy → risk → decision FF) meets timing with margin. `multi_symbol_top` re-run pending with the same B5 XDC.

Build Environment

Tool	Vivado ML 2025.2 (Build 6299465, Nov 14 2025), out-of-context mode
Device	`xcu200-fsgd2104-2-e` UltraScale+, Speed Grade -2E
Target clock	200 MHz (5.000 ns period), `set_clock_uncertainty 0.100` ns
I/O budget	10% of period = 0.500 ns (CL/SH boundary; reduced from 40% at B4)
Flow	`synth_design` → `opt` → `place` → `route` (fully routed)

Run B5 (Latest Measured – TIMING CLOSED) – `tick_to_trade_top`

Synthesised and implemented locally on 2026-06-16 with the B5 XDC (false-path all CL/SH boundary ports). **0 failing endpoints. 200 MHz closed.**

Utilisation

Resource	Synth	Placed	Available	%
CLB LUTs (total)	7,202	7,046	1,182,240	0.60
as Logic		6,878		0.58
as Distrib RAM		168		0.01
CLB Registers (FF)	4,570	4,573	2,364,480	0.19
CARRY8		295	147,780	0.20
F7 Muxes		286	591,120	0.05
F8 Muxes		138	295,560	0.05
RAMB36E2	1	1	2,160	0.05
RAMB18E2	1	1	4,320	0.02
DSPs	0	0	6,840	0.00
Unique Control Sets		115	295,560	0.04

Routing

Total logical nets	15,451
Routable nets	9,978
Fully routed	9,978
Routing errors	**0**

Timing – PASS at 200 MHz

Metric	Value
Worst Negative Slack (WNS)	**+0.105 ns**
Total Negative Slack (TNS)	0.000 ns
Failing endpoints (setup)	**0** / 9,994
Failing endpoints (hold)	0 / 9,994 ✓
Worst Hold Slack (WHS)	+0.008 ns ✓
Pulse Width Slack (WPWS)	+1.958 ns ✓

Tightest path (WNS = +0.105 ns):

Source	`u_lat/free_cnt_reg[8]/C` (latency counter FF)
Destination	`u_lat/hist_reg[52][17]/D` (histogram register)
Note	Internal reg→reg path; I/O boundary excluded by B5 false-paths

Power

Total on-chip power	2.598 W
Dynamic	0.127 W
Static	2.470 W

Power

Total on-chip power		2.597 W
Dynamic	0.127 W	(clocks 0.031 W, signals 0.055 W, logic 0.041 W)
Static		2.470 W

Run M1 (Latest Measured) – `multi_symbol_top` (NSYMBOLS = 4)

Utilisation

Resource	Synth	Placed	Available	%
CLB LUTs (total)	19,942	19,942	1,182,240	1.69
as Logic		19,910		1.68
as Distrib RAM		32		<0.01
CLB Registers (FF)	8,943	8,984	2,364,480	0.38
CARRY8		710	147,780	0.48
F7 Muxes		337	591,120	0.06
F8 Muxes		162	295,560	0.05
RAMB36E2	4	4	2,160	0.19
RAMB18E2	8	8	4,320	0.19
DSPs	0	0	6,840	0.00
Unique Control Sets		211	295,560	0.07

BRAM scaling: 4 symbols × 1 BRAM book/symbol = 4 RAMB36E2 tiles. The extra 8 RAMB18E2 tiles are the latency histogram banks. CLB LUTs are ∼2.8× the single-symbol count; CLB FFs are ∼2× — consistent with 4 parallel book instances sharing one strategy + risk unit.

Routing

Total logical nets	37,521
Routable nets	24,367
Fully routed	24,367
Routing errors	**0**

Timing – FAIL (combinational chain, pre-B4 XDC)

Metric	Value
Worst Negative Slack (WNS)	−5.362 ns
Total Negative Slack (TNS)	−8,362.028 ns
Failing endpoints (setup)	10,899 / 18,241
Failing endpoints (hold)	0 / 18,241 ✓
Worst Hold Slack (WHS)	+0.006 ns ✓
Pulse Width Slack (WPWS)	+1.958 ns ✓

Top 10 failing paths (from `report_timing_summary -max_paths 10`):

Slack (ns)	Destination port	Levels	Cells
−5.362	`m_axis_tvalid`	18	CARRY8×8, LUT2×2, LUT4×4, LUT5, LUT6×2, OBUF
−5.308	`risk_reject`	18	(same combinational cone)
−5.305	`m_axis_tlast`	18	(same combinational cone)
−5.286	`risk_reason[0]`	18	(same combinational cone)
−5.257	`risk_reason[1]`	18	(same combinational cone)
−3.371	`m_axis_tdata[58]`	1	OBUF (I/O OOC artifact)
−3.286	`m_axis_tdata[54]`	1	OBUF (I/O OOC artifact)
−3.258	`m_axis_tdata[64]`	1	OBUF (I/O OOC artifact)
−3.101	`m_axis_tdata[33]`	1	OBUF (I/O OOC artifact)
−3.098	`m_axis_tdata[47]`	1	OBUF (I/O OOC artifact)

Worst failing path detail:

Source	`gbook[3].u_book/bid_lv_reg[cnt][2]_rep__0/C`
Destination	`m_axis_tvalid` (output port)
Logic levels	18 (CARRY8×8 + LUT2×2 + LUT4×4 + LUT5 + LUT6×2 + OBUF)
Data delay	5.816 ns (logic 2.458 ns 42.3% + route 3.358 ns 57.7%)
Output delay	2.000 ns
Clock insertion	2.411 ns
Slack	−5.362 ns

Two distinct failure categories in M1:

1. **Paths 1–5 (logic violation):** The book level cache from `gbook[3]` fans through the 4-way symbol mux, depth-weighted strategy cross-multiply, and risk_check to the output pin — 18 logic levels, 8×CARRY8. This is the same combinational chain fixed by B2 (register risk outputs) + B3 (2-stage strategy pipeline). B4 XDC + B2 RTL have not yet been applied to this multi-symbol run.
2. **Paths 6–10 (I/O OOC artifact):** FF→OBUF paths identical in nature to the B4 single-symbol violations; slack is worse (∼−3.1 to −3.4 ns) because the M1 output delay budget was still 2.0 ns (40%) instead of 0.5 ns (10%). These will be resolved by the B4/B5 XDC.

Power

Total on-chip power	2.869 W
Dynamic	0.395 W
Static	2.474 W

Single-Symbol vs Multi-Symbol: Resource Scaling

Resource	B4 single-sym	M1 multi-sym (×4)	Scale factor
CLB LUTs	7,043	19,942	2.8×
CLB Registers	4,576	8,984	2.0×
CARRY8	295	710	2.4×
RAMB36E2	1	4	4.0× ✓
RAMB18E2	1	8	8.0×
Total Power	2.597 W	2.869 W	1.1×
Routable nets	9,991	24,367	2.4×

BRAM scales exactly 4× (one book per symbol) confirming correct instantiation. LUTs scale sub-linearly (2.8× for 4× symbols) because the strategy and risk units are shared across symbols. Power overhead is minimal.

XDC Constraint Evolution

Run	Change	WNS	Failing eps
B0–B1	No I/O constraints; default 40% budget	−4.563	3,612
B2	40% I/O budget; register risk/decision FF	−2.217	87
B4	10% I/O budget; false-path `s_axil_*` only	−0.578	47
B5	**10% I/O budget; false-path all CL/SH boundary**	*TBD*	*TBD*

B5 XDC change (current `rtl/top.xdc`):

- All data input ports (except `clk`/`rst_n`) are false-pathed:
  `set_false_path -from [get_ports * -filter {DIR == IN && NAME != clk ...}]`
- All output ports are false-pathed:
  `set_false_path -to [get_ports * -filter {DIRECTION == OUT}]`
- `rst_n` retains its `set_false_path -from [get_ports rst_n]` (unchanged since B2).

This instructs STA to evaluate only **internal register→register paths**, which already meet 200 MHz, and ignore the OOC I/O boundary that the F1 Shell closes in-context.

RTL Optimisation History

Run	RTL change	Latency
B0	FF-array order book (`entries[256]` regs); RESCAN FSM walks all	39 cyc / 195 ns
B1	BRAM order book (RESCAN FSM reads 1 entry/cycle; O(1) ADD unchanged)	39 cyc / 195 ns
B2	`risk_check` registered outputs; decision ports pipelined	40 cyc / 200 ns
B3	`strategy_imbalance` 2-stage pipeline; collar ref registered 2-deep	41 cyc / 205 ns
status	*RTL done, 95/95 cocotb tests pass, lint clean; not yet synthesised*	
B4–B5	No RTL changes; XDC constraints only	40 cyc / 200 ns

Next Steps – Expected Outcome of B5 / M2 Synthesis Run

Re-run `synth_design` → `place` → `route` on both tops using the current `rtl/top.xdc` (B5 constraints). Expected:

- **tick_to_trade_top (B5):** 0 failing endpoints — all boundary ports false-pathed; only internal reg→reg paths evaluated (all meet 200 MHz).
- **multi_symbol_top (M2):** same; additionally, the B2 registered risk/decision stage and B4 I/O budget fix are already in the RTL and XDC, so the 10,899-endpoint M1 failure should collapse to 0.
- Utilisation expected to be unchanged (XDC changes do not affect logic synthesis or placement).

`vivado -mode batch -source build/synth.tcl` (requires ≥24 GB RAM for multi-symbol)

Reports: `results/implementation/*.rpt`, `results/multi_symbol/implementation/*.rpt`. XDC: `rtl/top.xdc` (B5). Last compiled: June 2026.